

knowledge and skill-set of the relatively small dedicated group of Collabera Labs (which comprises a team of 40 developers), this platform taps into a resource pool of over 4000 employees of Collabera, working across the globe, for ideas, assets, and components that can add to the collective IP of the organisation. The aim is to improve productivity dramatically. While there are rewards and recognition programmes associated with this initiative (such as 'Most active participation', 'Most useful contribution', 'Most amount of reuse', etc), the company believes that the soft benefits of this initiative itself will attract contributors from across the company. "Once the initiative gains sufficient momentum, we also intend to make this part of individual KRAs (Key Result Areas); for example, participate in a Collabera open source initiative for a certain number of hours/day," says Variyam. At a practical level, the power of collaboration brings in scale. A 4000-strong developer community is 100 times bigger than the 40-member innovation group, he adds.

Driven by open source technologies

Given the ultimate intent of the initiative, Collabera Labs has built the entire platform based on open source technologies, states Variyam. "We have used the Python-based Trac as the project- and source-management platform. Its powerful Web-based interface also provides an integrated wiki and issue/defect tracking features. Trac also has an interface to Subversion, which is the underlying source-control management tool. Subversion provides plug-ins for all popular IDEs, such as Eclipse and Visual Studio. This makes the entire platform accessible to all developers, without any need for additional software and tools. The Apache HTTP server is used as the Web server to host the platform, which also provides integration to LDAP for authentication."

A novel way to develop software and innovate

Since the most important ingredient required for the success of such an initiative is collaboration, organisations that want to adopt this model need to encourage their employees to share and solicit ideas and participation, emphasises Variyam. This means that openness is not just in the source code and other engineering aspects, but also in terms of being open to peer reviews and constructive criticism. There needs to be a realisation across the organisation that the end goal is to create an asset that's larger than the sum of all individual contributions. This will help everyone in the organisation strive towards that common goal, says Variyam.

"The key thought behind this model is to ensure that every project and every individual contributes to the development and creation of assets that can be used and reused multiple times. This is a good way to improve productivity and the quality of work, while maintaining competitive advantage," he adds.

The way the open source innovation model works

The model comprises a process that revolves around three primary 'actors' and four key steps for each of the hosted projects. The 'actors' include:

- *The project owner*, who manages a specific project and sets up the required environment and repositories for that project. This person is also responsible for ensuring that various steps in the open source development life-cycle are adhered to.
- *The contributor*: This person could be anyone in the company, who plays the role of a developer, reviewer, or tester. A contributor works collaboratively with the rest of the contributors and the project owner.
- *The consumers are those who want to use assets from a particular project in their own area of development.* The consumers also raise issues and suggest possible enhancements for the assets, and create the feedback mechanism that is essential for the continuous improvement of all assets.

Now, moving on to the four key steps:

- The first phase begins from the *Idea Space* where all ideas are born. This is a discussion forum where anyone can create/share an idea that has the potential to be taken up as an open source project. Everyone chips in, and after the pros and cons are discussed and debated, some of these ideas will get into the open source platform as projects. This is the ideation and initiation phase. At this point, a project owner is identified.
- During the next phase of *collaborative development and review*, contributors can start collaborating on this project as developers, reviewers, or testers. During the review phase, apart from ensuring code quality, extra focus is given to make sure that there are no IP infringements.
- During the *integration* phase, source code and assets from various contributors are integrated into a deployable build, which is then tested as a single unit. Issues are logged, fixed, and the build re-tested.
- The *implementation and improvement* phase closes the loop, and provides feedback to the entire lifecycle. During this phase, consumers look for reusable assets built as part of any of the open source projects. Improvement opportunities are identified, and a new iteration of a project starts.

At any point in time, there could be multiple projects and associated assets at various stages of the lifecycle. Every project can have a set of stable versions and an alpha/beta version available.

But is it practically viable?

It would be interesting to explore how viable this model is, at a practical level. Could it lead to a scenario where too many cooks end up spoiling the broth? Variyam disagrees: "The power of crowd-sourcing and peer review is tremendous, and becomes, in itself, a self-correcting/optimising mechanism.